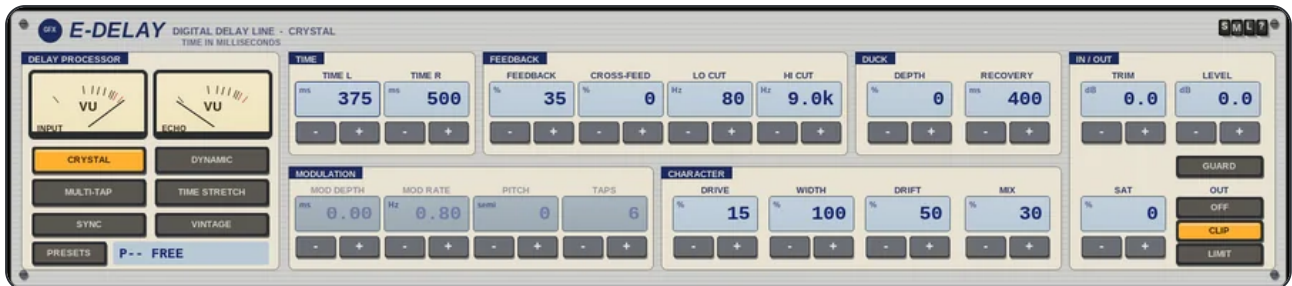


E-Delay

Four European studio and broadcast delays of the 1980s: pristine repeats, ducking, multi-tap and pitch loops.



Overview

E-Delay brings together four digital delays in the spirit of the European studio and broadcast units of the 1980s: clean, precise repeats, careful spatial design and transformer-coupled inputs. The MODEL keys pick the algorithm.

CRYSTAL is an ultra-clean dual delay with feedback filters and a cross-feed for ping-pong; when you change the time it crossfades to the new one, so the pitch never bends. DYNAMIC is the same line with a ducker: the repeats dip while you play and bloom back when you stop. MULTI-TAP spreads up to eight taps across the time, alternating sides, each gently chorused. TIME STRETCH runs the feedback through a pitch shifter and a soft limiter, for endless rising or falling repeats that never run away.

Left and right times are set separately in milliseconds or synced to the project tempo. The DUCK section works on every model, and VINTAGE runs the line through the converters of the era.

Reach for it for

- Clean slapback and tempo-synced echoes on vocals and guitars
- Ping-pong delays with the cross-feed
- Vocal echoes that stay out of the way while the singer sings (DUCK)
- Rhythmic multi-tap patterns and chorus-like taps
- Shimmering, rising or falling pitch-shifted repeats (TIME STRETCH)

Controls

Working the displays

Every parameter has its own display. Drag up or down on a display (Shift for fine moves), use the mouse wheel over it, or press the - / + keys underneath (hold for repeat). Double-click a display to reset it to its default. The last display you touched is outlined. Displays that do nothing on the current model are dimmed.

DELAY PROCESSOR

INPUT / ECHO meters	VU meters showing the input level and the level of the echoes (0 VU = -18 dBFS).
CRYSTAL	The ultra-clean dual delay: feedback filters, cross-feed for ping-pong, and a crossfade when the time changes, so the pitch never bends.
DYNAMIC	The same line with the ducker working. Choosing it sets DUCK DEPTH to 60% if it was at 0.
MULTI-TAP	Up to 8 taps (TAPS) spread evenly across the longer of TIME L and TIME R, alternating left and right, each quieter than the last and gently chorused by MOD DEPTH and MOD RATE. The feedback comes from the end of the line.
TIME STRETCH	The repeats pass through a pitch shifter (PITCH, -12 to +12 semitones per repeat) and a soft limiter inside the loop, so they can rise or fall forever without running away.
SYNC	Locks TIME L and TIME R to the project tempo; the displays then show note values (1/1 down to 1/32, with dotted D and triplet T values) and the header shows the tempo.
VINTAGE	Runs the delay line through 16-bit converters at the era's clock rate: 48 kHz on CRYSTAL, 32 kHz on DYNAMIC and MULTI-TAP, 40 kHz on TIME STRETCH.
PRESETS	Opens the list of factory presets.
◀ / ▶	Step to the previous or next preset (wrapping round).

Preset display

Shows the last preset recalled (P1 - P10, or P--) and whether the times are FREE or SYNC.

TIME

TIME L / TIME R

The left and right delay times, 1 to 2700 ms (375 and 500 ms by default), or note values when SYNC is on (1/8D and 1/4 by default).

FEEDBACK

FEEDBACK

How much of the echo goes back into the line, 0 to 99% (35% by default). More gives more repeats.

CROSS-FEED

How much of each side's feedback crosses to the other side, 0 to 100%. Turn it up for ping-pong echoes.

LO CUT

A high-pass filter on the repeats, 20 to 1000 Hz (80 Hz by default). Each repeat gets thinner.

HI CUT

A low-pass filter on the repeats, 1 to 20 kHz (9 kHz by default). Each repeat gets darker.

DUCK

DEPTH

How far the echoes are pulled down while you play, 0 to 100% (up to about 18 dB). Works on every model.

RECOVERY

How long the echoes take to bloom back when you stop, 20 to 2000 ms (400 ms by default).

MODULATION

MOD DEPTH

How far each tap's time is swept, 0 to 10 ms. MULTI-TAP only.

MOD RATE

The speed of that sweep, 0.05 to 10 Hz. MULTI-TAP only.

PITCH

The pitch shift applied to each repeat, -12 to +12 semitones. TIME STRETCH only.

TAPS

The number of taps, 1 to 8 (6 by default). MULTI-TAP only.

CHARACTER

DRIVE

The input transformer, 0 to 100% (15% by default). The lows saturate first, for warmth and weight before the delay.

WIDTH

The stereo width of the echoes, 0 to 150%. 0 is mono; above 100 is wider than normal.

DRIFT

How much the delay clock wanders, 0 to 100% (50% by default). See Drift below.

MIX

The dry / echo balance, 0 to 100% (30% by default).

Drift

The delay clock is not quite steady: the left and right times each wander slowly and independently by a fraction of a millisecond, which keeps the repeats from sounding sterile. At 50% you get the standard amount; 100% doubles it and makes it wander a little faster; 0% switches it off.

IN / OUT

TRIM

Input level before anything else, -24 to +12 dB.

LEVEL

The output level, -24 to +12 dB.

GUARD

Off by default. When on, it gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. GUARD CATCHING shows while it is working.

SAT

Band saturation on the way out, 0 to 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.

OFF / CLIP / LIMIT

The final stage: nothing, a soft clipper at -0.3 dBFS (no latency), or a lookahead limiter that holds the peaks at -0.3

dBFS (3 ms latency, compensated by REAPER). The OUT caption turns into a red CLIP while the clipper is working.

The header

S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

Presets set the model, times, sync, feedback section, modulation, duck, taps and mix. Drive, vintage, width, drift and the IN / OUT section are left as they are, except Reset.

Slapback	CRYSTAL: 95 / 110 ms, one quiet repeat.
Quarter Note Echo	CRYSTAL synced to quarter notes, 35% feedback.
Ping-Pong Eighths	CRYSTAL synced to 1/8 and 1/4 with 70% cross-feed for ping-pong.
Dynamic Vocal Echo	DYNAMIC: dotted eighth and quarter, ducked by 70% while the vocal sings.
Wide Stereo Doubler	CRYSTAL: 17 / 29 ms with no feedback, for width.
Rhythm Taps	MULTI-TAP: six taps spread across a quarter note (synced), lightly modulated.
Tap Chorus	MULTI-TAP: four taps across 120 ms with 3 ms of modulation, for a chorus-like spread.
Shimmer Up	TIME STRETCH: each repeat an octave up, 70% feedback, a little duck.
Falling Echoes	TIME STRETCH: each repeat 5 semitones down.

Endless Wash	TIME STRETCH: 97% feedback with no pitch shift: near-endless repeats.
Reset all to defaults	Every control back to its starting value, including the IN / OUT section.

Tips

- For vocals, try DUCK DEPTH at 50 - 70%: the echoes stay out of the way of the words and fill the gaps.
- Lower HI CUT and raise LO CUT for repeats that sit behind the mix rather than on top of it.
- In TIME STRETCH, +12 or +7 semitones with high feedback makes a shimmer; keep MIX low and HI CUT down to tame it.
- On a send, set MIX to 100% and use LEVEL to set the return level.

File: gfx-e-delay.jsfx